

Information Theory Behind Jujutsu Kaisen: Gojo Satoru's Unlimited Void



Amirlan Kalmukhan

Essays on the Science of Fiction, No. 1

September 2026

1 Introduction

On the B5F platform of Shibuya Station, boxed in by four special grade curses and about a thousand transfigured humans, Gojo Satoru opens his Domain Expansion and closes it again after 0.2 seconds.

The number is not there for drama. Akutami states it, and gives Gojo a reason for it. Two tenths of a second was Gojo's own estimate of how long an ordinary person could stand inside Unlimited Void and walk away without lasting damage. It works. Everyone on the platform, curse and civilian alike, is left unconscious but still upright, which buys Gojo the 299 seconds he needs to destroy the transfigured humans one by one.

Then the manga adds a detail, and that detail is the reason this essay exists. In that fifth of a second, the people caught inside were exposed to about half a year's worth of information.



Figure 1: The Shibuya platform in the moment after Unlimited Void closes. Civilians, curses and transfigured humans are all unconscious and all still standing. Nobody has been hit by anything. Image source: Jujutsu Kaisen Wiki (Fandom), *Unlimited Void*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

That sentence is doing something unusual for a manga. It is not describing an effect, it is quoting a rate: six months of experience, divided by two tenths of a second. As far as I can tell it is the only hard number in the entire series that says what Unlimited Void actually does to a brain. And it can be checked, because how much information a person can absorb per second is not a mystery. People have been measuring it since 1952.

What follows takes that number seriously. The main tool is information theory, which Claude Shannon invented in 1948 while thinking about telephone lines, and which turns out to describe Gojo's domain far better than it has any business doing. The

answers are more specific than I expected when I started. If Akutami's figures are right, the rate Unlimited Void has to deliver lands within 27 percent of the measured bandwidth of the human nervous system. The moment of failure can be pinned down to about 38 billionths of a second. And the technique's entire useful range, from the dose a civilian survives to the dose that permanently damages Ryomen Sukuna, is a factor of fifty.

The argument also ends up being slightly rude about the technique's reputation. Unlimited Void is not powerful. Measured against what 1.4 kilograms of matter can physically compute, it is roughly forty-one orders of magnitude too weak to be worth anyone's attention. What it is instead is very precisely aimed, and by the end it should be clear why that is the more frightening property of the two.

2 What the Technique Actually Does

Before doing anything with that number, here is exactly what the source material claims. The claims are unusually specific for a shonen manga, and three of them carry real weight.

Unlimited Void (無量空処, *Muryōkūsho*) first appears in chapter 15, against the special grade curse Jogo. The domain drops the target inside the Limitless, and boundless raw information floods into their mind. What the manga then says about the experience is the first important claim: the target can see and feel everything while not being able to see or feel anything at the same time. Every living process they have, perception and communication included, is forced to work infinitely, and they can do nothing but die slowly. Gojo's own summary, delivered while Jogo stands there unable to assemble a thought, is that the target has been given everything and still cannot do anything at all.



Figure 2: Jogo inside Unlimited Void in chapter 15. He is conscious, physically unharmed, and completely unable to act. The manga is clear that the incapacitation is informational rather than physical, and that Gojo shut the domain before Jogo actually died. Image source: Jujutsu Kaisen Wiki (Fandom), *Unlimited Void*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

The second claim is about delivery. A finished Domain Expansion seals the caster’s technique inside a barrier, and anything cast within that barrier is guaranteed to hit. Chapter 15 is specific that this guaranteed hit gets past even Gojo’s own Infinity, a defence that works purely on distance. So whatever Unlimited Void does to a target, it does not do it by firing something across a room.

The third is the Shibuya figure already mentioned: 0.2 seconds, about half a year of information, picked as the largest dose Gojo thought a non-sorcerer could take without permanent harm.

Two smaller pieces of the setting matter later on. Kento Nanami, sizing the technique up in chapter 30, treats it as an almost guaranteed win once it catches someone, which is a strong thing to say in a series where nearly everything has a counter. And Kinji Hakari’s domain, in chapter 187, has a guaranteed hit whose entire payload is the transfer of information to its target. Information as a weapon is already a mechanic in this world. It is not something being smuggled in from outside.

Now back to that first claim. Someone who perceives everything and nothing at the same moment sounds like a writer reaching for an image. It is not. It is a theorem, and it was proved in 1948.

3 Shannon and the Idea of a Channel

In 1948 a mathematician at Bell Telephone Laboratories named Claude Shannon published a paper about a very practical problem. Given a telephone line, how much can

you push down it before the pushing stops being worth anything? His answer applies to every channel that has ever existed, telephone wires and optic nerves alike.

Shannon's first move was to make information countable. He defined the information in a message by how surprising it is. A message you could have predicted tells you nothing; a message you could not tells you a lot. The unit is the bit, and one bit is the amount of information in the answer to a single well-posed yes-or-no question. Everything after that is bookkeeping.

His second move is the one that matters here. Every channel, Shannon showed, has a single number attached to it. He called it the capacity, wrote it C , and measured it in bits per second. Capacity is the fastest you can send information through that channel and still have it arrive intact.

Then comes the theorem, and it has two halves. Below capacity, you can encode your message cleverly enough to get it through essentially perfectly, however noisy the line is. Above capacity, you cannot get it through at all.

That second half is the surprising one, because the obvious guess is wrong. You might reasonably expect that transmitting twice as fast as a channel allows would get you half your message through, and transmitting a thousand times too fast would get you a thousandth. It does not work like that. Past capacity the failure is close to total almost immediately. Shannon's converse theorem puts a hard floor under the damage:

$$P_{\text{error}} \geq 1 - \frac{C}{R},$$

where R is the rate you are actually transmitting at. Send at ten times capacity and at least 90 percent of the message is lost. Send at a million times capacity and 99.9999 percent is gone.

Stated plainly, and this is the whole essay in one sentence: an overloaded channel does not hand you a garbled, partial version of the message. It hands you nothing. Above capacity there is no partial credit.

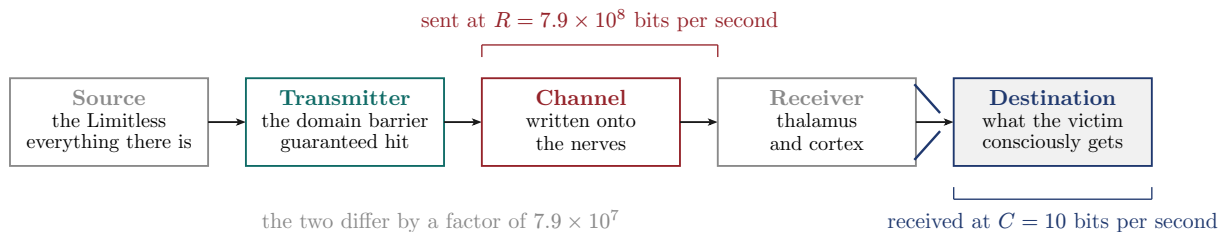


Figure 3: Unlimited Void drawn as the communication system Shannon sketched in 1948. Every link is ordinary until the last one, where a channel carrying 7.9×10^8 bits per second arrives at a destination that can accept ten. Shannon's original diagram has a box for noise, left out here because the technique does not need any. The signal is perfectly clean. The victim fails anyway, and the narrowing at the final junction is the entire reason.

Which takes us straight back to chapter 15. The manga says that a victim of Unlimited Void can see and feel everything while not being able to see or feel anything

at the same time. That is not a paradox and it is not a flourish. It is exactly what Shannon's theorem predicts. The signal arrives in full, which is the seeing and feeling everything. None of it can be decoded, which is the seeing and feeling nothing. Both halves are true simultaneously, and they have to be.

I doubt Gege Akutami derived the converse to the coding theorem. What I think happened is that he tried to describe honestly what infinite input would feel like from the inside, and ended up where Shannon ended up with algebra. Of the two possibilities that is the more interesting one.

4 How Much a Human Can Actually Receive

Everything now depends on the size of C for a person, and that is a measurement rather than an argument. People have been chipping away at it for seventy years, using methods that have almost nothing in common, and the answers keep landing in the same place.

Start with 1952. A British psychologist named W. E. Hick sat people in front of a set of lamps and asked them to press the matching key when one lit up. The interesting part was not how fast they reacted but how their speed changed as he added lamps. Reaction time did not double when he doubled the alternatives. It grew with the logarithm of the number of choices, which is precisely how the amount of information in a choice grows. Hick had caught human decision making obeying Shannon's arithmetic four years after Shannon published it. Read as a rate, his result comes to roughly 6.7 bits per second. Ray Hyman replicated it the following year from a different direction, by varying how likely each signal was rather than how many there were, and got the same law.

Jump to 2019. Christophe Coupé and colleagues took fifteen texts, had them translated into seventeen languages across nine families, and recorded native speakers reading them. Languages differ enormously in speed. Japanese speakers push out about eight syllables a second, Thai speakers closer to five. But Japanese syllables carry less, and Thai syllables carry more, and the two effects cancel almost exactly. Every language in the study delivered close to 39 bits per second. Whatever sets that number, it is not the language. It is the listener.

The most thorough estimate is the most recent. In 2025 Jieyu Zheng and Markus Meister published a survey in *Neuron* that went looking for the ceiling everywhere they could think of: typing, reading, speech, competitive video games, memory championships, blindfolded Rubik's cube solving. The tasks have nothing in common except a human doing them as fast as possible. The answer came back at about 10 bits per second.

Ten bits per second is a humbling number. It is roughly the rate at which you can type. A home Wi-Fi connection moves five million times more than that.

The same paper supplies the other half of the story, and the other half is where the trouble starts. Your sensory organs collect data at something like 10^9 bits per second. Between what your senses gather and what you actually experience there is a factor of a hundred million, and essentially none of it survives the crossing. The optic nerve alone carries about 10^7 bits per second, a figure Koch and colleagues obtained in 2006 by measuring the output of a retina directly. You discard almost all of it before you ever know it existed.

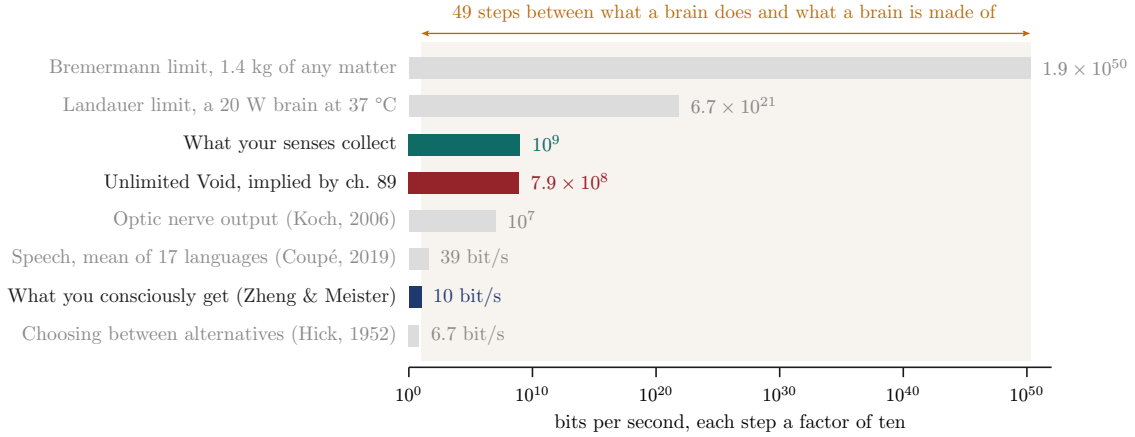


Figure 4: Human information rates set against the physical limits of the matter a brain is built from. Each tick is a factor of ten, and the chart spans fifty-two of them. Two things stand out. The rate Unlimited Void must deliver to match Akutami’s own numbers (red) sits almost on top of the rate your senses actually collect (green). And the shaded strip is the gap this essay keeps returning to: a brain does its conscious work forty-nine factors of ten below what its own matter could manage. Unlimited Void operates inside that strip and nowhere near its ceiling.

One last way to feel the size of this. Half a year of conscious experience, at ten bits per second, works out to 1.578×10^8 bits. That is under twenty megabytes. Six months of everything you noticed, chose, felt and said, stored at the rate you actually live it, would fit in about five phone photographs.

5 The Arithmetic of Half a Year

Now the calculation the manga makes possible. Take Akutami at his word: an exposure of $T = 0.2$ seconds carrying half a year of information. Half a year is 1.578×10^7 seconds, and at ten bits per second that is

$$I = 1.578 \times 10^7 \text{ s} \times 10 \text{ bit/s} = 1.578 \times 10^8 \text{ bits.}$$

To deliver that much in two tenths of a second, the domain has to run at

$$R = \frac{I}{T} = \frac{1.578 \times 10^8 \text{ bits}}{0.2 \text{ s}} = 7.89 \times 10^8 \text{ bits per second.}$$

The measured figure for how fast information moves up the human nervous system is 10^9 bits per second. The two agree to within 27 percent.

A what the domain compresses

■ exposure: 0.2 seconds



B the rate that requires, against rates we have measured

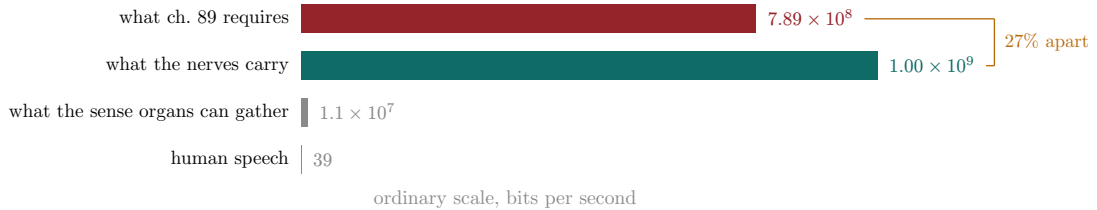


Figure 5: The central calculation. Panel A shows the compression the manga specifies: two tenths of a second carrying 182.6 days of conscious experience. Panel B turns that into a delivery rate and compares it, on an ordinary scale rather than a logarithmic one, with rates that have actually been measured in people. What the fiction demands and what the biology provides differ by 27 percent. The third bar matters too: the sense organs are seventy-two times too slow to carry this, so the information cannot be arriving through them.

Two honest caveats before going further. Akutami did not do this calculation, and I am not suggesting he did. Neither of his numbers was chosen with Shannon in mind. It is entirely possible the agreement is a fluke, and two numbers plus one ratio is thin evidence for anything.

But there is a reason it might not be a fluke, and it does not require the author to know any information theory at all. What Akutami needed was a dose that felt like the absolute maximum a person could be given and still wake up. A dose at the maximum a person can take is, by definition, a dose that fills the receiving hardware completely. Filling the hardware completely means running at its bandwidth. So the storyteller's instinct for "as much as a human being could possibly be given" and the laboratory's measurement of "as much as a human being can possibly receive" are pointed at the same physical quantity. Landing within 27 percent of each other is less miraculous than it first looks. It is still a good showing for a weekly manga.

The calculation also produces something the manga never says, and this is the part I find genuinely satisfying. Add up the capacity of every sense organ in the body, using Manfred Zimmermann's classic accounting, and you get about 1.1×10^7 bits per second, with the eyes contributing roughly nine tenths of it. Unlimited Void needs 7.89×10^8 . It requires seventy-two times more bandwidth than every sense you own put together can supply.

So the information cannot be coming in through the eyes and ears. There is no room for it. It has to be written directly onto the nerves behind them.

That is a conclusion drawn from nothing but division, and the manga confirms it independently. Chapter 15 establishes that a completed domain's guaranteed hit bypasses spatial defences, Infinity included. Something that ignores distance is not something crossing a room and entering an eye. The arithmetic and the lore arrive at

the same architecture from opposite ends, which is a better result than this kind of exercise usually gets.

6 How Fast It Happens

With both rates fixed, the converse theorem stops being abstract. The overdrive is

$$\frac{R}{C} = \frac{7.89 \times 10^8}{10} = 7.89 \times 10^7,$$

and Shannon's floor gives

$$P_{\text{error}} \geq 1 - \frac{C}{R} = 1 - 1.27 \times 10^{-8} = 0.999999987.$$

This is not a claim that understanding becomes difficult. It is a proof that understanding is unavailable. No clever encoding, no training, no quantity of cursed energy and no amount of being Ryomen Sukuna moves this bound, because the bound contains no term for how good the receiver is. It contains only capacity.

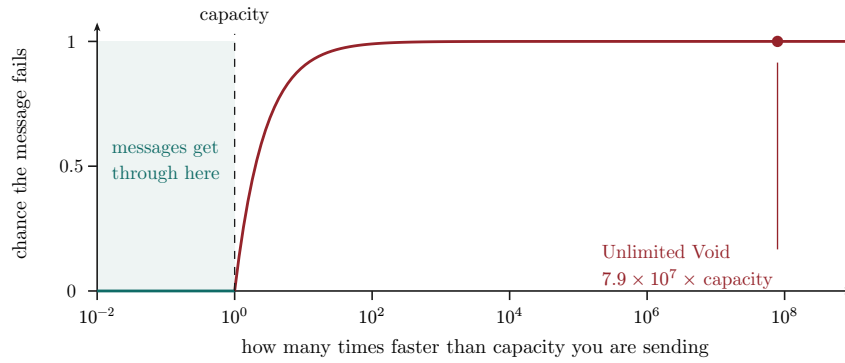


Figure 6: Shannon's cliff. To the left of the dashed line you are sending slower than the channel allows, and with enough cleverness the message arrives perfectly. To the right there is no cleverness available: the failure rate climbs the curve $1 - C/R$ and flattens against certainty. What the picture rules out is the intuitive middle ground, where sending too fast gets you a degraded but partly useful message. That region does not exist. Unlimited Void sits almost eight steps out along the bottom axis.

Now the timing, which is what the premise of this essay actually asked for. Think of the victim as a clerk with an in-tray. Work arrives at rate R , the clerk processes it at rate C , and the in-tray holds B items before it overflows. The in-tray here is working memory, and the standard estimates put it at about four items, or equivalently two to four seconds' worth of conscious throughput. Call it 30 bits. It fills in

$$t_{\text{overflow}} = \frac{B}{R - C} = \frac{30}{7.89 \times 10^8} = 3.8 \times 10^{-8} \text{ s} = 38 \text{ nanoseconds}$$

Thirty-eight nanoseconds is the answer, and it should be admitted straight away that it is the answer to a slightly different question than a neuroscientist would have asked. Nothing biological happens in 38 nanoseconds. A nerve impulse takes about a millisecond, some twenty-six thousand times longer. Crossing a single synapse takes

half a millisecond. In 38 nanoseconds light travels 11.4 metres, roughly the width of the platform Gojo is standing on.

So the honest statement is not that the brain fails after 38 nanoseconds. It is that the overflow is not an event the nervous system is fast enough to take part in. By the time the first neuron involved has finished firing once, working memory has been exceeded about twenty-six thousand times over. The failure is not a process with a duration. It is the starting condition.

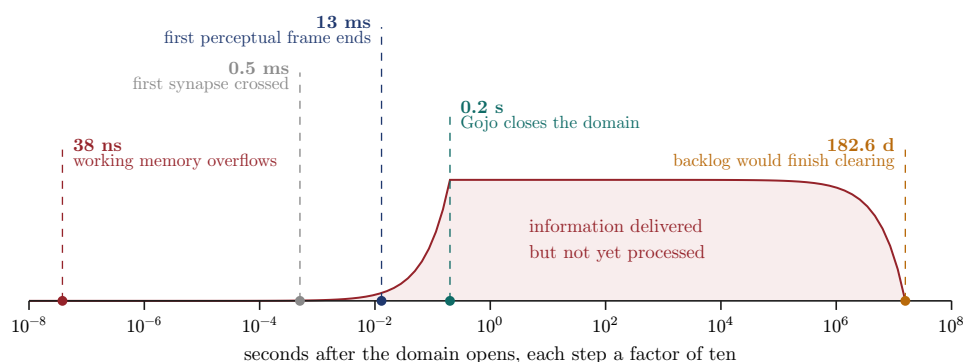


Figure 7: The whole event on one axis, from billionths of a second to months. The shaded shape is the pile of information that has been delivered but not yet dealt with. It passes the capacity of working memory 38 nanoseconds in, thirteen thousand times before a single synapse has even been crossed, and it is still growing when Gojo shuts the domain at two tenths of a second. Clearing that pile at ten bits per second would take 182.6 days, which is the half year the manga names. The dose and the recovery turn out to be the same quantity written in different units.

The timeline also explains something about Gojo's tactics that is easy to miss. Conscious perception is not continuous, it is sampled, and Mary Potter's group showed in 2014 that a picture flashed for as little as 13 milliseconds can still be understood. That puts one frame of conscious experience somewhere around that length. In the first such frame, Unlimited Void delivers about ten million bits and the victim can accept 0.13 of one. They cannot finish acquiring a single complete bit until a tenth of a second has passed, by which point seventy-nine million bits are waiting.

So the incapacitation is finished inside the first frame. Everything after that adds to the pile rather than doing fresh harm, which means the technique's output grows in a straight line with time rather than exploding. That is exactly why 0.2 seconds is a dial Gojo can set at all. He is not balancing on a cliff edge. He is choosing how many frames to charge his victims for, and he chose fifteen.

7 Why the Body Stops

The manga is insistent about one particular symptom. Victims do not collapse, convulse or bleed. They stand there. Jogo stays conscious and lucid enough to notice that he is helpless, and cannot act on the observation.

Hick's law explains this exactly, using nothing beyond the 1952 paper. Response time grows with the amount of information in the choice being made,

$$RT = a + b \log_2(N + 1), \quad b \approx 0.15 \text{ seconds per bit},$$

where N is the number of options. Because the growth is logarithmic, going from two choices to four costs the same extra time as going from four to eight. The law is really a statement that choosing is rate-limited, at roughly seven bits per second.

Now let N run off to infinity. The logarithm diverges, and the response time diverges with it. A muscle command comes after a decision, the decision never finishes, and so no command is ever sent. The body is not paralysed. The body is waiting.

Put Akutami's dose into the formula. A victim handed 1.578×10^8 bits of choice needs

$$RT = 0.15 \times 1.578 \times 10^8 = 2.37 \times 10^7 \text{ s} \approx 274 \text{ days}$$

to make one decision. Gojo's line in chapter 15, that the target has been given everything and still cannot do anything at all, is not an observation about irony. It is Hick's law evaluated at the limit. The people standing motionless on the Shibuya platform have not been frozen. They are still choosing.

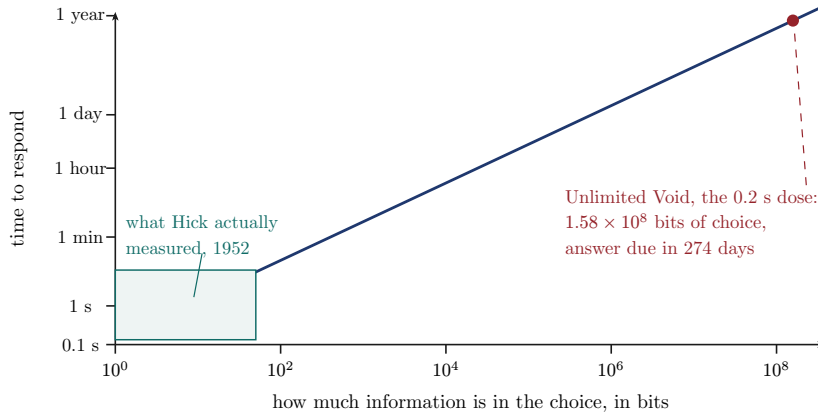


Figure 8: Hick's law, extended far past where anyone has tested it. The straight line is the 1952 result, measured over a range of one to three bits (shaded box) and never seriously overturned since. Unlimited Void presents 1.58×10^8 bits of choice. The line does not bend, break or level off there. It simply keeps going, and the answer comes due nine months late. Worth noting on the side: Hick's slope implies a decision rate of 6.7 bits per second, within a factor of 1.5 of the 10 bits per second Zheng and Meister arrived at in 2025 by completely unrelated means, seventy-three years later.

8 Why Nothing Actually Burns

The manga says victims of Unlimited Void die slowly, and that Gojo closed the domain before Jogo perished. Death by this technique is real, then, but indirect. Something that cannot choose an action cannot defend itself, cannot leave, and eventually cannot keep itself alive. The information is lethal the way a locked door is lethal.

That still leaves the question of whether something else is doing the work. "The brain overloads and burns out" is the intuitive story, and it happens to be wrong in three separate ways.

The first check is heat. In 1961 Rolf Landauer showed that erasing information has an unavoidable thermodynamic price, about 3×10^{-21} joules per bit at body temperature. Erasing Akutami's entire six-month dose therefore costs

$$1.578 \times 10^8 \text{ bits} \times 2.97 \times 10^{-21} \text{ J/bit} = 4.7 \times 10^{-13} \text{ joules.}$$

For comparison, one nerve impulse in one cortical neuron costs roughly 1.9×10^{-11} joules. The whole half year of information, thermodynamically destroyed, comes to about one fortieth of the energy of a single neuron firing a single time. Against the brain's 20 watt budget over those two tenths of a second, it is one part in ten trillion. Nothing overheats and nothing burns.

The second check is storage. There is a hard ceiling, found by Jacob Bekenstein in 1981, on how much information can be packed into a region of a given size and energy. For something the size and mass of a human head it comes to about 10^{43} bits. Unlimited Void, running flat out, would need roughly 5×10^{26} years to approach it, which is some thirty-eight million billion times the current age of the universe. The skull is in no danger of running out of room.

The third check is the one that reframes everything else. In 1962 Hans Bremermann derived the fastest rate at which any matter whatsoever can process information, about 10^{50} bits per second per kilogram. For a 1.4 kilogram brain that is 1.9×10^{50} bits per second. Unlimited Void delivers 7.9×10^8 . The attack is forty-one orders of magnitude too slow to trouble the matter it is aimed at. A perfectly engineered 1.4 kilogram computer would swallow the whole thing and not notice.

That leaves exactly one place for the harm to live. A brain does its conscious business forty-nine factors of ten below the limit of the stuff it is made from, and Unlimited Void is pointed straight into that margin. This is not a strong technique. It is an accurate one, and the difference matters: strength would have to beat physics, while accuracy only has to find the gap between what a system is built from and what it actually does with itself. That gap is the largest undefended margin in human biology, and we all walk around inside it.

It also means the phrase this essay started from, cognitive burnout, is the wrong model. Nothing burns. The channel jams. That distinction is not pedantry, because burning is permanent and jamming is not, and the manga is consistent with jamming throughout. Jogo recovers. The Shibuya civilians recover. The one character who does not recover is the one who refuses to throw anything away.

9 Sukuna and the Compressibility of Everything

Over the Shinjuku fight Gojo expands Unlimited Void five times, across chapters 225 to 229. In the fifth clash Sukuna is late opening Malevolent Shrine by under 0.01 seconds, which is long enough to hold him still for a moment, and a moment is long enough for Gojo to land a severe strike. Added up, the manga puts Sukuna's total

exposure at under ten seconds, and the result is brain damage on a par with Gojo's own, bad enough that Malevolent Shrine is unavailable for the rest of the fight.



Figure 9: Sukuna caught inside Unlimited Void in chapter 229. The characters on the panel, 無量空処, name a Buddhist meditative state rather than a weapon, which the conclusion comes back to. Image source: Jujutsu Kaisen Wiki (Fandom), *Unlimited Void*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Set that beside the civilian number. An ordinary person handles 0.2 seconds. The strongest being in the series handles something under ten. That is a factor of fifty, against a technique that overdrives its target by 7.9×10^7 . Fifty is not a defence, it is a rounding error on eight orders of magnitude. This is what Nanami was getting at in chapter 30 when he called the domain an almost guaranteed win. Across the entire range of biological and supernatural variation the series contains, Unlimited Void does not care who you are. Cursed energy buys strength, speed and durability, and it does not appear to buy bandwidth.

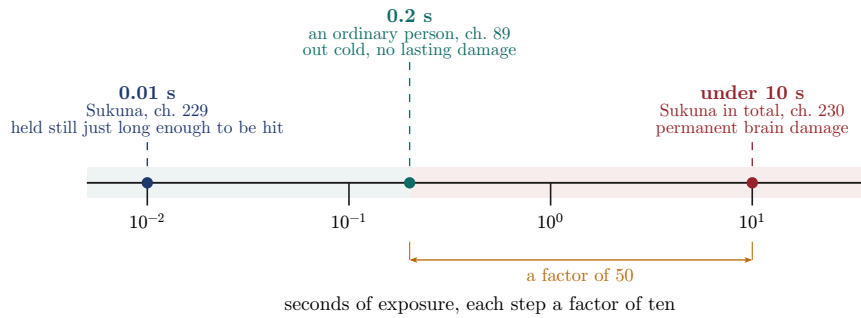


Figure 10: Every dose the manga actually puts a number on, laid out on one axis. The technique’s whole working range, from the exposure that buys a single opening against Sukuna to the exposure that permanently damages him, covers three factors of ten. The distance between an ordinary human’s tolerance and the King of Curses’ is a factor of fifty. Against an overdrive of 7.9×10^7 , that difference does not decide who survives. It only decides how long the sentence takes to carry out.

So how does anyone beat it? The manga’s answer is better than a power-level argument, and it is consistent every single time.

Mahoraga adapts, and Sukuna arranges for Megumi Fushiguro’s soul to take the brunt of the first five domains so the adaptation can finish. In Shannon’s language, adapting to a source is compression. Compression works by exploiting predictability: if you know that the letter Q is nearly always followed by U, you need barely any bits to record the U. A receiver that does not yet know the pattern has to reserve space for every possibility, which is enormously wasteful. Learning the pattern is what reclaims the waste. Mahoraga’s turning wheel is a codebook being assembled.

That it converges in five exposures, under ten seconds in total, says something specific about what is being sent. The source must be hugely predictable. And it would be. Unlimited Void shows the target everything, and everything is deeply patterned. A description of the universe is vastly longer than the rules that generate it. Against an infinite stream of genuinely random noise there would be no pattern to find and nothing to adapt to, and Mahoraga would have turned its wheel forever. The technique’s one structural weakness is that it tells the truth, and the truth compresses.

Which brings up the detail I keep coming back to. There are four successful responses to Unlimited Void in the manga. Sukuna touches Gojo to invoke the contact exemption in chapter 227. Malevolent Shrine cuts at the domain’s barrier from outside in chapter 225. Mahoraga destroys the domain outright in chapter 229. Megumi’s soul is made to receive the load in Sukuna’s place in chapter 230. Two of those destroy the sender and two of them refuse to be the receiver. Not one of them involves processing the information faster. In a series otherwise deeply committed to the idea that enough talent and preparation can overcome anything, nobody ever out-thinks this technique, because the rule that governs it has no term for how clever you are.

10 Conclusion

Take Akutami at his word and the technique resolves into something unexpectedly precise. Unlimited Void writes information onto the nervous system at about 7.9×10^8 bits per second, which is within 27 percent of what that system is measured to carry, and seventy-two times more than the sense organs could ever deliver, so the payload has to bypass them exactly as the guaranteed-hit mechanic requires. The conscious receiver on the far end runs at ten bits per second. The resulting mismatch of 7.9×10^7 puts the failure rate at $1 - 1.3 \times 10^{-8}$ by a theorem that admits no exceptions. Working memory is formally exceeded 38 nanoseconds in, before any neuron involved has finished a single impulse, and the incapacitation is complete inside the first frame of conscious perception at around 13 milliseconds. The paralysis is Hick's law with the number of options sent to infinity. And the six months of recovery is the same six months of information, divided by the speed at which a person is able to live.

None of the damage is energetic. The full dose costs a fortieth of one nerve impulse to erase, and runs forty-one factors of ten below what the brain's own matter could handle. The attack works because a brain does vastly less computing than it is physically capable of, and Unlimited Void is aimed into precisely that shortfall.

There are limits to all of this and they should be said plainly rather than buried in a footnote. Cursed energy appears in no equation here and obeys no conservation law anyone has written down. "About half a year's worth of information" is a line in a weekly manga, not a measurement, and the fact that it divides neatly into Zheng and Meister's bandwidth figure may well be luck. Working memory is not really a 30-bit in-tray, and the queue model that produces 38 nanoseconds is a cartoon of a nervous system. What survives all of that is the shape of the argument rather than its decimal places. Any receiver of finite capacity, given an unlimited rate, fails completely and at once, and no property of that receiver except its capacity enters the calculation.

There is one last thing I cannot let go of. *Muryōkūsho* is not a word Akutami made up. It is the Japanese form of *Ākāśānantyāyatana*, the sphere of infinite space, the first of the four formless absorptions in Buddhist cosmology: a meditative state in which the mind passes beyond all perception of form and matter. Unbounded extent, and no structure anywhere inside it. That is a fifth-century description of a place with infinite information and nothing whatsoever to understand, attached fifteen hundred years later to a technique whose entire function is to transmit everything and communicate nothing. Shannon would have recognised the address.

Because what Shannon really proved is that information is not a substance and cannot crush anybody. It is a relationship between a message and whoever is receiving it, and that relationship can fail on only one side. Gojo does not overwhelm anyone. He transmits accurately, at the rate the universe is willing to hand him, to somebody who listens at ten bits per second. Nothing in that exchange is unfair, and nothing in it can be survived by being stronger.

References

- [1] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, vol. 27, pp. 379–423 and 623–656, 1948.
- [2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006, chs. 2, 7 and 8.
- [3] J. Zheng and M. Meister, “The unbearable slowness of being: Why do we live at 10 bits/s?,” *Neuron*, vol. 113, no. 2, pp. 192–204, 2025. doi:10.1016/j.neuron.2024.11.008
- [4] W. E. Hick, “On the rate of gain of information,” *Quarterly Journal of Experimental Psychology*, vol. 4, no. 1, pp. 11–26, 1952.
- [5] R. Hyman, “Stimulus information as a determinant of reaction time,” *Journal of Experimental Psychology*, vol. 45, no. 3, pp. 188–196, 1953.
- [6] G. A. Miller, “The magical number seven, plus or minus two,” *Psychological Review*, vol. 63, no. 2, pp. 81–97, 1956.
- [7] N. Cowan, “The magical number 4 in short-term memory,” *Behavioral and Brain Sciences*, vol. 24, no. 1, pp. 87–114, 2001.
- [8] C. Coupé, Y. M. Oh, D. Dediu and F. Pellegrino, “Different languages, similar encoding efficiency: Comparable information rates across the human communicative niche,” *Science Advances*, vol. 5, no. 9, eaaw2594, 2019.
- [9] K. Koch, J. McLean, R. Segev, M. A. Freed, M. J. Berry II, V. Balasubramanian and P. Sterling, “How much the eye tells the brain,” *Current Biology*, vol. 16, no. 14, pp. 1428–1434, 2006.
- [10] M. Zimmermann, “The Nervous System in the Context of Information Theory,” in *Human Physiology*, R. F. Schmidt and G. Thews, Eds., 2nd ed., Springer, 1989, pp. 166–173.
- [11] M. C. Potter, B. Wyble, C. E. Hagmann and E. S. McCourt, “Detecting meaning in RSVP at 13 ms per picture,” *Attention, Perception & Psychophysics*, vol. 76, no. 2, pp. 270–279, 2014.
- [12] R. Landauer, “Irreversibility and heat generation in the computing process,” *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183–191, 1961.
- [13] J. D. Bekenstein, “Universal upper bound on the entropy-to-energy ratio for bounded systems,” *Physical Review D*, vol. 23, no. 2, pp. 287–298, 1981.
- [14] H. J. Bremermann, “Optimization through evolution and recombination,” in *Self-Organizing Systems*, M. C. Yovits, G. T. Jacobi and G. D. Goldstein, Eds., Spartan Books, 1962, pp. 93–106.
- [15] D. Attwell and S. B. Laughlin, “An energy budget for signaling in the grey matter of the brain,” *Journal of Cerebral Blood Flow & Metabolism*, vol. 21, no. 10, pp. 1133–1145, 2001.
- [16] A. D. Baddeley, *Working Memory*, Oxford University Press, 1986.
- [17] G. Akutami, *Jujutsu Kaisen*, Shueisha, 2018–2024. Chapters 8, 15, 30, 84, 89, 119, 164, 187 and 225–230.
- [18] Jujutsu Kaisen Wiki, *Unlimited Void*, https://jujutsu-kaisen.fandom.com/wiki/Unlimited_Void, retrieved September 2026.
- [19] Jujutsu Kaisen Wiki, *Domain Expansion*, https://jujutsu-kaisen.fandom.com/wiki/Domain_Expansion, retrieved September 2026.